QCert: Verifying Sixteen Quantum Circuit Completeness Results in One Agda Framework

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A set of circuit equations is complete if any two circuits denoting the same linear map can be derived from each other using these equations, and minimal if no equation can be derived from the others. Establishing these properties takes long case analyses spanning hundreds of pages, making the details hard to check by hand. To this end, we extend the previous formalization framework to QCert, and use it to formalize the main circuit completeness theorems to date. This includes sixteen results, fifteen on completeness and one on minimality, over finite gate sets such as Clifford, CNOT-dihedral, Clifford+T, Clifford+CS, real Clifford+CH, and Toffoli–Hadamard, on qubits and on qudits of every odd prime dimension. QCert divides the work between humans and LLM coding agents. We first built its framework by specifying the syntax, semantics, and group presentations. Then, each paper's main theorem can be expressed uniformly in this framework. Next, an agent took over and formalized the proofs in Agda, following the papers' arguments. Agda checked every proof, so a human need only check the relation table, the gate semantics, and the theorem statement for each paper. Overall, QCert confirms the main results of each work and identifies gaps in two of the existing proofs. Its ability to do verification at scale makes QCert well suited to verifying quantum circuit completeness in higher dimensions.

1 Introduction

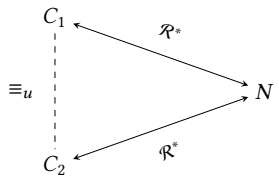
An equational theory for a family of quantum circuits is a set of equations between circuits used as rewrite rules. By applying these rules to subcircuits, we can transform a circuit into an equivalent one. Such a theory should be *sound* because every rewrite preserves the underlying linear map. It should also be *complete* so that any equivalence between circuits can be derived. Additionally, we want it to be *minimal*, with no redundant rules. Equational rewriting supports circuit optimization [26, 32, 33, 40], equivalence checking [41], and hardware mapping [38]. Complete theories also support reasoning about quantum programs. For example, Hadamard-Π [21] is a complete equational theory for reversible quantum programs. Completeness results have been established for a wide range of circuit families, including reversible linear circuits [27, 35], reversible circuits over multiple-control Toffoli gates [22, 23], Clifford [7, 28, 31, 37], CNOT-dihedral [1, 12], two-qubit Clifford+T [9], three-qubit Clifford+CS [10], Toffoli–Hadamard [2] and real Clifford+CH [16] circuits, and with continuous parameters, all quantum circuits [18, 19]. Completeness has also been obtained for the group of two-level matrices underlying several of these circuit families [8, 15, 21, 24, 29]. Minimal theories have been given for six of these circuit fragments [13] and for all quantum circuits [17].

There are two main strategies for proving completeness of a rule set $\mathcal{R}$ over a gate set $\mathcal{G}$. Our machine-checked framework builds on both of them, so we start by sketching

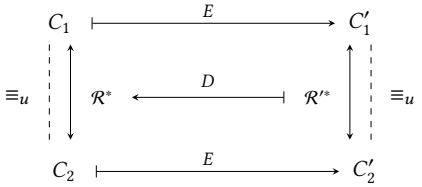
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the ideas behind them. Consider any two circuits C_1 and C_2 over $\mathcal{G}$ that have the same linear map, $C_1 \equiv_u C_2$. We need to show that C_1 can be rewritten into C_2 using $\mathcal{R}$. Figure 1a outlines the first strategy. It specifies a unique normal form for each linear map [1, 7, 12, 22, 23, 27, 28, 31, 35, 37], often by giving an exact synthesis algorithm [8, 15, 21, 24, 29]. Completeness follows by proving that every circuit can be transformed into its normal form using the rewrite rules. By uniqueness, C_1 and C_2 have the same normal form N. Reversing the path from C_2 to N and concatenating it with the path from C_1 to N gives

$$C_1 \overset{\mathcal{R}^*}{\longleftrightarrow} N \overset{\mathcal{R}^*}{\longleftrightarrow} C_2.$$



(a) The rewrites from C_1 and C_2 to their common normal form N using $\mathcal{R}$ give rewrites from C_1 to C_2 using $\mathcal{R}$.



(b) Completeness of $\mathcal{R}'$ gives rewrites between $C_1' = E(C_1)$ and $C_2' = E(C_2)$. Translating this proof back gives rewrites between C_1 and C_2 using $\mathcal{R}$.

Fig. 1. Two strategies for proving completeness of $\mathcal{R}$ over a gate set $\mathcal{G}$. $\overset{\mathcal{R}^*}{\longleftrightarrow}$ means applying a finite sequence of rewrites using $\mathcal{R}$ in either direction, where $*$ denotes zero or more steps.

Figure 1b illustrates the second strategy, which uses an already complete rule set $\mathcal{R}'$ over a different gate set $\mathcal{G}'$ [2, 9–11, 13, 16, 18–20, 36]. Here, E and D translate circuits from $\mathcal{G}$ to $\mathcal{G}'$ and back, replacing each gate with a circuit over the other gate set. Both E and D preserve circuit composition and the underlying linear map. To transfer completeness from $\mathcal{R}'$ to $\mathcal{R}$, we first show that every rule of $\mathcal{R}'$, translated by D, can be derived using $\mathcal{R}$. We can then replace each step of the proof between C_1' and C_2' by its translated derivation, giving $D(C_1') \overset{\mathcal{R}^*}{\longleftrightarrow} D(C_2')$. We also need to check that each gate $g \in \mathcal{G}$ can be rewritten to its round-trip translation: $g \overset{\mathcal{R}^*}{\longleftrightarrow} D(E(g))$. Applying these rewrites gate by gate gives $C_j \overset{\mathcal{R}^*}{\longleftrightarrow} D(C_j')$ for $j \in \{1, 2\}$. Combining these paths yields

$$C_1 \overset{\mathcal{R}^*}{\longleftrightarrow} D(C_1') \overset{\mathcal{R}^*}{\longleftrightarrow} D(C_2') \overset{\mathcal{R}^*}{\longleftrightarrow} C_2.$$

Sometimes the two strategies are combined, as in the scaled dyadic extension of [29].

Starting from a complete rule set, we can remove a redundant rule by deriving it from the other rules, thereby preserving completeness. To show that the resulting rule set is minimal, we need to give, for each remaining rule R, an interpretation of circuits that every other rule preserves but R does not. Any derivation using the other rules preserves this interpretation, so R cannot be derived from them [13, 17].

Completeness and minimality proofs typically require long case distinctions over gates, relations, and normal forms, whose supporting derivations sometimes span hundreds of pages [16, 21, 28]. In particular, relation reductions require intricate circuit manipulations and repeated rewrites, making them prone to human mistakes. For example, circular

reasoning can lead us to remove a rule whose derivation still relies on that rule, hidden in a chain of other derivations. Errors can then propagate through dependent derivations, invalidating the entire reduction and making the error source hard to trace by hand.

Our contributions. To make completeness proofs robust and manageable as they extend to new gate sets, we turn to a machine-checked framework, where a proof assistant like Agda [14, 34] verifies every derivation and its dependencies. A human then only needs to check that the framework is set up correctly and the main theorems are stated as intended. Consistent definitions of circuits and their matrices also allow us to compare different completeness results in a unified setting. Because all proofs are checked at scale, such a framework paves the way for verifying circuit completeness in higher dimensions.

This work builds on our formalization of qudit Clifford completeness [7], where we built the framework [4] and carried out the main verification steps by hand. LLM coding agents came in only at the final stage, to connect various steps into the formal proof. Our experience is part of a broader shift where LLMs are now used to write large formal proofs. Anthropic, for example, reports that LLMs wrote a 13-million-line Lean proof of Fermat's Last Theorem within 11 days [3]. In a hybrid approach, humans write and carefully review the definitions and theorem statements [25], coding agents write the proofs, and Agda checks them. Following this, we extend our previous framework [4] to QCert and use it to formalize completeness and minimality results from sixteen papers, as shown in Table 1.

In Section 2, we describe how QCert adds matrix semantics to our earlier Agda library of circuits and presentations [4], so that every completeness theorem is stated consistently as a group presentation. In Section 3, we present our formalization workflow and the techniques that keep typechecking fast. Then in Section 4, we demonstrate the usefulness of QCert by machine-checking the completeness and minimality results of sixteen papers. In Section 5, we briefly describe the issues these formalizations found in the source papers. In Section 6, we discuss the current limitations of QCert and our plans to extend it.

2 The QCert Framework

QCert states every theorem as a group presentation. A circuit is a word over an alphabet X of gates, and the rules form a relation Γ on words, which generates a congruence $\approx$. A semantics is a homomorphism $[\![-]\!]$ from words to a group G, such as a matrix group. Soundness and completeness say the relations present the image of $[\![-]\!]$, which is the subgroup of G generated by the gates. Our earlier library [4] expresses this as a record. In Agda, mathematical structures such as groups are defined as records. The record Γ IsSubPresentationOf G contains the semantics $[\![-]\!]$, with proofs of soundness and completeness. The record Γ IsPresentationOf G adds surjectivity, so the relations present all of G.

Unlike the letters of an ordinary word, gates act on wires. The gates in X are indexed by the number of wires they act on. Wires are numbered from 0 at the bottom, and a gate always acts on the bottom wires of a circuit. To place it anywhere else, we apply a shift, which takes a circuit on n wires and places it on wires $1, \ldots, n$ of a circuit with $n + 1$ wires. Each relation is stated on as many wires as the paper draws it on. To apply it inside a circuit with more wires, we need *structural rules*, which hold for every circuit, regardless of its

gate set. These rules say that an equation can be shifted up by one wire, gates on disjoint wires commute, and a scalar can be placed on any wire. They are intuitive to humans and thus rarely stated, but a formalization in Agda must define them explicitly.

QCert then gives circuits a matrix semantics $[\![-]\!]$ with entries in one of the rings $\mathbb{Z}\left[1/\sqrt{2}, i\right]$, $\mathbb{Z}\left[1/2, i\right]$, $\mathbb{Z}\left[1/\sqrt{2}\right]$, and $\mathbb{Z}\left[1/2\right]$, which are used in exact synthesis algorithms. The target group G is the unitary group over that ring. It is built from the group structures of Agda's standard library and defined once per ring, so it is checked once. As matrices, a gate on the bottom wires and a shift are Kronecker products with an identity, so the extension of $[\![-]\!]$ from gates to circuits is defined once for all gate sets. Since the structural rules are universal, their soundness is also proved only once. Then for each paper, it suffices to check the soundness of its own rules.

The formalization of each paper ends with its main theorem stated as a record of one of these two kinds, IsSubPresentationOf or IsPresentationOf. For example, Statement (1) is the theorem for the real stabilizer circuits of [31], with the relations R1–R16 of its Figure 1 lifted to n wires. Statement (2) is the theorem for the two-level matrices of Hadamard-Π [21], with the relations of its Figure 6.

```
presentation : {n : ℕ} → (n VRel,_===_) IsSubPresentationOf U (2 ^ n)    (1)
presentation : {n : ℕ} → (_===_ {n}) IsPresentationOf (O n)              (2)
```

Agda checks every proof, so a reader only needs to check that each theorem statement matches its paper's result. Since every theorem is one of these two records, the reader checks only four components: the relations transcribed from the paper, the target group, the matrix of each gate, and the line assigning $[\![-]\!]$ to the record. All sixteen statements are collected in a single root module, so the reader can check them in one place.

3 Formalization Workflow

Our workflow has three parts. First, before formalizing a paper, we check its rewrite rules, its intermediate results, and each line of its derivations with exact matrices in short scripts. This caught most typos before any Agda code was written. Second, we state each main theorem in QCert before a coding agent starts, so the agent only supplies the proof. Finally, for long derivations, the agent writes a script that turns each line of the paper into one step of the Agda proof, so a failing step points to a specific line of the paper.

The main computational cost came from Agda's conversion checker. If two terms are syntactically identical (i.e., written in the same way), it accepts them immediately. Otherwise, it evaluates both, which can take exponential time for a long matrix product. For such products, we keep circuit interpretations abstract and decide matrix equality by computing a single Boolean using row operations, which is orders of magnitude faster than comparing matrices entry by entry. We record these techniques in the instructions followed by the agents, so later proofs avoid the same slowdowns.

4 Formalized Completeness and Minimality Proofs

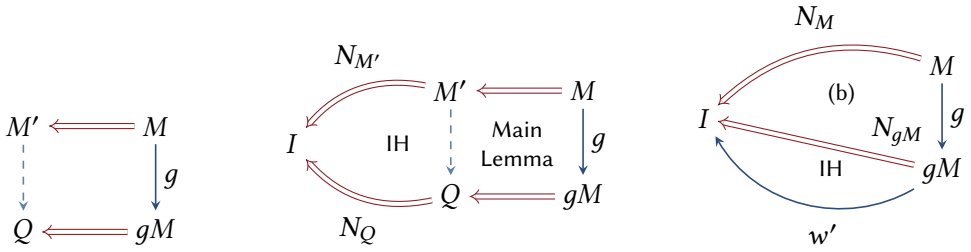
The sixteen formalizations in Table 1 contain about 420,000 lines of Agda code and use no postulates, and each theorem is as general as the result in its source paper. Moreover, all matrices are computed exactly. Their entries lie in algebraic number rings, whose elements

are given by finitely many integers, so Agda checks automatically whether two matrices are equal. Over the real or complex numbers, no such automatic check exists, which limits the equality checking of QCert to discrete gate sets, as we discuss in Section 6.

Source	Object	Result	Mod	Time
Groups of m-level matrices				
Greylyn [2014]	$U_n(\mathbb{Z}[1/\sqrt{2}, i])$	presentation, $n \leq 4$	31	42
Bian and Selinger [2021]	$U_n(\mathbb{Z}[1/2, i])$	presentation	37	62
Li et al. [2021]	$O_n(\mathbb{Z}[1/2])$, L_n	presentations	68	79
Fang et al. [2026]	$O_n(\mathbb{Z}[1/\sqrt{2}])$	presentation	44	44
Clément [2019]	$O_n(\mathbb{Z}[1/\sqrt{2}])$	presentation	40	64
Circuits				
Selinger [2015]	Clifford	sub-presentation	31	34
Bian, Li et al. [2026c]	Clifford, odd prime p	presentation of a model	215	546
Makary et al. [2021]	real stabilizer	sub-presentation	30	35
Amy et al. [2018]	CNOT-dihedral	sub-presentation	30	24
Blake [2026b]	phase-affine, prime d	sub-presentations	56	61
Bian and Selinger [2023a]	two-qubit Clifford+T	sub-presentation in U_4	62	118
Bian and Selinger [2023b]	three-qubit Clifford+CS	sub-presentation in U_8	123	250
Clément [2026]	real Clifford+CH	sub-presentation	346	405
Amy et al. [2024]	three-qubit Toffoli–Hadamard	presentation of L_8	201	102
Li et al. [2025]	qutrit Clifford	presentation	89	282
Minimality				
Blake [2026c]	six fragments	minimal, complete	73	183
Total			1,476	2,332

Table 1. Completeness and minimality results formalized in QCert. A presentation gives generators and relations for a whole group, and a sub-presentation for the subgroup generated by the gate set named in the Object column of the Circuits block. L_n is the group of orthogonal matrices $N/\sqrt{2}^j$, where N is an integer matrix and $j \geq 0$. Mod denotes the number of Agda modules, the units that Agda checks one at a time. Time records their typechecking time in seconds from scratch, measured with Agda 2.8 on an Intel Core i5-14400. Mod and Time consider only each row's own modules, not the shared library or the formalizations it builds on.

The completeness proofs follow the two strategies of Figure 1. For the groups of two-level matrices, the normal forms come from exact synthesis. Each matrix, or *state*, has a *level*, built from quantities such as the least denominator exponent of a column, and the algorithm reduces each state M to the identity in steps that lower the level. These steps form the normal word N_M. The *Main Lemma* says that for every state M and generator g, a diagram of relations closes the square formed by g and the algorithm's next steps from M and gM, passing only through states of lower level (Figure 2a). By well-founded induction on the level, the relations then rewrite $N_{gM}\, g$ into N_M (Figure 2b), and by induction on length, every word that reduces M to the identity rewrites into N_M (Figure 2c). Two words with the same matrix therefore rewrite into each other. We formalize these inductions once and prove the Main Lemma for each presentation.



(a) The Main Lemma. (b) Induction on the level. (c) Induction on length.

Fig. 2. Completeness from the Main Lemma. Plain arrows are words, double arrows are steps or runs of the algorithm, and dashed arrows are words through states of lower level.

Most theorems in Table 1 are proved from scratch in QCert against a matrix semantics, with three exceptions. The two rows of [9, 10] build on the Agda proofs that accompany those papers. These proofs relate the circuit relations to two-level relations by the Reidemeister–Schreier method [30]. We connect them to our formalizations of the two-level matrix presentations in [8, 24], the first two rows of the table, to obtain sub-presentations of $U_4\left(\mathbb{Z}\left[1/\sqrt{2}, i\right]\right)$ and $U_8\left(\mathbb{Z}\left[1/2, i\right]\right)$. The qudit Clifford theorem of [7] is the only one whose semantics is not a matrix group. Its group presentation is obtained by lifting a presentation of the symplectic group $\mathrm{Sp}(2n, \mathbb{Z}_p)$ with Pauli corrections and global phases, and its semantics is built in the same way.

5 Corrections to the Source Papers

The formalizations in Table 1 identify two kinds of issues in the source papers. Most are typos, and some of them appear in statements that the proofs rely on. These are easy to fix. Two others are harder to repair. One is a gap in a proof, and the other is a relation missing from a theorem. We describe both below.

The gap is in the proof of Theorem 4.14 of Hadamard-Π [21], which states that the relations in its Figure 6 present the group $O_n\left(\mathbb{Z}\left[1/\sqrt{2}\right]\right)$. One case of the proof relies on Lemmas A.16 and A.17, which have counterexamples in dimension 6, and the case analysis built on them misses a case in dimension 9. The authors have confirmed this gap (personal communication). Clément's unpublished notes [15] present the same group with different relations, and his case analysis covers the missing case, using which the formalization of Theorem 4.14 is completed. We also prove that the two sets of relations are equivalent in every dimension and formalize Clément's proof [5].

The missing relation is in Theorem 5.2 of [2]. It states that the relations in its Figures 4 and 7, over the Toffoli and $\mathrm{K} = \mathrm{H} \otimes \mathrm{H}$ gates, present the group $O_8\left(\mathbb{Z}\left[1/2\right]\right)$. This is the first of two steps towards presenting the scaled dyadic orthogonal group L_8 by three-qubit Toffoli–Hadamard circuits. The relation r_{K} in Section 5.5, $\mathrm{K}_{1,2} = \mathrm{K}_{[4,5,6,7]}\,\mathrm{X}_0\,\mathrm{K}_{[4,5,6,7]}\,\mathrm{X}_0$, is in neither figure. Nor can it be derived from them. Hence, Theorem 5.2 fails as stated, and we have informed the authors. Our formalization [6] shows that r_{K} is independent by giving an interpretation of circuits that every other relation preserves except r_{K}, and proves the presentations of $O_8\left(\mathbb{Z}\left[1/2\right]\right)$ and L_8 with r_{K} restored.

6 Discussion and Future Work

We formalize the results of sixteen papers in QCert, using consistent circuit syntax, a unified definition of group presentations, and shared libraries for rings and their matrix groups. Since all formalizations use these definitions, presentations of the same group from different papers can be compared and proved equivalent. The same libraries, together with the induction that turns an exact synthesis algorithm into a completeness proof, can be reused for new algorithms. Only the case analysis over matrices and generators is specific to each algorithm. As the most error-prone part of the proof, it is where machine checking in QCert matters most.

At the moment, not every completeness result fits into QCert, as its scope is limited in two ways. First, it supports only discrete gate sets, so it does not cover completeness results for circuits with continuous parameters, such as rotations by arbitrary angles [17–19]. Second, control is not a circuit constructor in QCert, so it does not cover results that treat control as a primitive operation on circuits [11, 20, 36].

We plan to extend the infrastructure of QCert to more families of quantum circuits, including the two above. Completeness is still open for many discrete gate sets in higher-dimensional quantum systems, and their case analyses are often too large to check by hand. In particular, we plan to study the completeness of Clifford circuits in all dimensions with the help of QCert. Beyond completeness, the exact synthesis libraries can be reused for T-count optimization of multi-qubit Clifford+T circuits. Finally, we are translating the formalizations from Agda into Lean, to make them available to its larger community. This will also allow us to take advantage of Lean's rich mathematical library Mathlib [39].

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